
Artificial Intelligence, Intellectual Property and Criminal Liability: Exploring the role of AI in technology-driven crimes and the legal challenges of accountability

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ABSTRACT:

The rapid advancement of Artificial Intelligence (AI) has transformed innovation, automation, and intellectual property (IP) creation while simultaneously facilitating technology-driven crimes such as deep fakes, cyber fraud, automated piracy, and IP infringement. Existing criminal law frameworks, which primarily rely on the principles of *actus reus* and *mens rea*, face significant challenges in attributing criminal liability when offences involve autonomous or semi-autonomous AI systems. This study examines the legal accountability of AI in technology-related offences, particularly those involving intellectual property violations, and explores the liability of developers, users, deployers, and corporations in the absence of AI legal personality. Adopting a doctrinal, comparative, and analytical research methodology, the study analyses statutory provisions, judicial decisions, policy frameworks, and scholarly literature from India, the European Union, the United States, and the United Kingdom. The findings reveal that existing criminal and intellectual property laws are inadequate to address the complex distribution of responsibility arising from AI-enabled offences. Comparative analysis demonstrates that while several jurisdictions have introduced regulatory measures to govern AI, significant accountability gaps remain due to the absence of harmonized legal standards. The study proposes a hybrid liability framework incorporating elements of strict and vicarious liability, algorithmic transparency, audit mechanisms, explain ability, and human oversight to strengthen accountability. It concludes that an adaptive and internationally coordinated legal framework is essential to balance technological innovation with effective criminal enforcement and intellectual property protection in the evolving AI ecosystem.

Keywords: Artificial Intelligence, Intellectual Property, Technology-related offences. Criminal Liability, Intellectual Property, AI Accountability

1. Introduction

John McCarthy, who is regarded as '*the father of Artificial intelligence*' has defined Artificial Intelligence(AI) as 'the science and engineering of making intelligent machines, especially intelligent computer programs'¹. In other words, AI can be defined as the capability of a machine to imitate intelligent human behavior by way of data that is feed into it.

AI can be categorized into two main types: narrow and general AI. Narrow AI which is also called weak AI is the AI that is built to perform certain specific functions and which we use on daily basis, examples may include voice assistant etc. General AI which is also known as strong AI can perform across various domains and can outperform humans.²

With the rise of usage of AI worldwide, the AI is also increasingly being used in criminal activities like deep-fake, cybercrime, hacking, digital arrest etc. Due to AI, it is difficult to trace criminal and attribute liability which is a major challenge. Federal Bureau of Investigation of the USA in its 2025 report observed that around 20 billion dollars were defrauded by using AI to commit cybercrimes in the USA.³ Reports also suggest that AI- enabled ransomware attacks reached record levels in 2025-2026, which were used in automated phishing attacks, identification of system vulnerabilities by automated AI attacks⁴. AI is also widely used in deep-fake pornography targeting women, children.⁵

In a recent case of Ahmedabad, India, AI was used by the criminals in identity theft to manipulate Aadhar-linked data. The accused generated fake facial blink videos from photographs to bypass KYC verification. They got access to Digi locker, banking data , financial data.⁶ Recently, Shashi Tharoor has approached Delhi High Court seeking protection against deep-fake videos that allegedly violated his personality rights.⁷ Many high-profile celebrities have approached Courts to protect them against misuse of their personality Rights by deep-fakes, voice cloning, identity theft. An important development in this regard has come from the US where thousands of authors alleged that Anthropic AI has used their protected work in training their Claude AI model. The authors sued them in a US court and Anthropic has agreed to settle the dispute with the author by paying them 1.5 billion Dollars.⁸ This case can open up a Pandora's box as there are multiple such cases which are pending across different jurisdictions.

Criminals used AI to generate deep-fake video and voice technology to impersonate a senior executive of a firm named Arup. Employees transferred around 25 million dollars during a conference meeting thinking the fake video as a real person.⁹ Google reported that for the first time hackers have used AI to detect vulnerability in software which was unknown for decades. This exploit could be used in autonomous

¹ John McCarthy, *What is Artificial Intelligence?* (Stanford Univ. 2007)

² Wim Naude & Nicola Dimitri, *The Race for an Artificial General Intelligence: Implications for Public Policy*, 35 AI & Soc'y 367 (2020)

³ <https://www.fbi.gov/news/press-releases/cryptocurrency-and-ai-scams-bilk-americans-of-billions> , last visited on 10/05/2026

⁴ <https://www.cybersecuritydive.com/news/cybercrime-ai-ransomware-mcp-malwarebytes/811360/> , last visited on 10/05/2026

⁵ Max Kamachee, Stephen Casper, Michelle L. Ding, Rui-Jie Yew, Anka Reuel, Stella Biderman & Dylan Hadfield-Menell, *Video Deepfake Abuse: How Company Choices Predictably Shape Misuse Patterns*, arXiv:2512.11815[cs.CY] (2025)

⁶ <https://timesofindia.indiatimes.com/city/ahmedabad/ai-enabled-aadhar-fraud-racket-busted-four-arrested/articleshow/130589941.cms> , last visited on 10/05/2026

⁷ <https://timesofindia.indiatimes.com/india/Shashi-tharoor-moves-delhi-high-court-block-ai-deepfakes/articleshow/130968876.cms> , last visited on 15/05/2026

⁸ <https://www.reuters.com/legal/government/us-judge-considers-anthropics-15-billion-settlement-authors-lawsuit-2026-05-14> , last visited on 10/05/2026

⁹ <https://www.weforum.org/stories/2025/02/deepfake-ai-cybercrime-arup/> last visited on 10/05/2026

cyberattacks and cybercrimes.¹⁰ AI is used in a large scale in committing crimes but the liability of AI is uncertain. Who would be held liable when AI is used in commission of the crime. This is global concern as there is no dedicated framework or institution to govern AI, neither AI has a legal personality so it can't be made liable for the crimes committed. This research work primarily focuses on the liability aspect of AI when AI is being used to commit offences, discusses the present regulatory framework in India and world over and brings out possible measures to bring AI under the regulatory framework.

Elements of Criminal Liability

The fundamental principle of criminal liability is *actus-reus* and *mens-rea*. *Mens-rea* is criminal intent to commit an offence. This criminal intent could be intention of doing an offence, knowledge that the offence could be committed by the act, or it could be negligence or recklessness. These varying degrees of criminal intent is what decides the extent of culpability of an accused. The problem lies in the fact that artificial intelligence as an independent entity lacks consciousness, as of now, and therefore is unable to form guilty intent by itself. It is either by developers, deployers or some other entities that AI can be misused. So the question of making AI criminally liable by itself is a problematic proposition as of today. In almost every jurisdiction, AI has not been assigned any legal personality and therefore in cases of autonomous AI carrying out cybercrimes, it can't be held liable. A major proposition of this paper is to assign legal personality to AI, especially autonomous AI so as to make it liable like corporations.

Since hundreds of years, Corporations were not subject to Criminal Law in accordance with the Roman Law *Societas Delinquere Non Potest* meaning Corporation cannot perform delinquency. It was only in the seventeenth century that the English Courts¹¹ started to impose criminal liability on the corporation which remain to the present day. A question arises as to why AI should also be not treated in a similar manner as corporations.

Fixing liability of AI

Gabriel Hallevy¹² in his work focused on a fundamental question as to whether the AI is accountable when AI is used for criminal acts. His argument lies in the fact that Criminal act requires *mens rea* and since AI lacks consciousness and therefore guilty mind, therefore it will not be enough to hold AI accountable. He develops a '*Perpetration via another*' model according to which AI is just an instrument and the real perpetrator would be the individual who uses AI to commit such act. He also advocated for a '*Direct liability model*' in which the AI could be held liable if the AI model is capable of operating autonomously.

There is a consensus that while fixing liability of AI, instead of making every sort of AI personally liable for the offences committed using it, there could be a breakdown of the types of AI and fixing liability on the basis of the degree of autonomy and intellect the AI has of its own. Accordingly, AI could be divided into 3 categories:

- When the AI is acting as a mere intermediary and it is totally dependent on the command of the developer or end-user, in such a scenario the developer or end-user could be held liable for the offences committed using AI.

¹⁰ <https://www.reuters.com/legal/litigation/hackers-pushing-innovation-ai-enabled-hacking-operations-google-says-2026-05-11>, last visited on 10/05/2026

¹¹ Case of Langforth Bridge, (1634) 79 Eng. Rep. 919 (K.B.)

¹² Gabriel Hallevy, *The Criminal Liability of Artificial Intelligence Entities-from Science Fiction to legal Social control*, 4 Akrin Intell. Prop. J. 171, 174-78(2010)

- When the AI is semi-autonomous entity, in that scenario also the developer or end-user could be held liable for the acts of AI.
- The third type is futuristic in nature which is autonomous AI where AI is no longer dependent on experiences or observations to learn and it has cognitive capabilities. In such a scenario AI could be made criminally liable.

*How to 'punish' an AI*¹³: Generally criminal offences carry punishments such as capital punishment, imprisonment, fine etc. The issue with AI is how to carry out such punishment when AI is a robot or a software. The possible solutions can be that for AI, the death sentence could be deletion of the software controlling the AI entity. For sentence of imprisonment, the operation of AI could be suspended for a definite period. Community sentencing could also be imposed on the AI where the AI could work for welfare of the community for dedicated time period.

Accountability of AI

There are AI which function as '*Black boxes*' i.e. it is very difficult to ascertain as to how they arrive at a particular decision. This lack of transparency also leads to difficulty in ascertaining liability for AI related offences. In this context, it is important to devise mechanisms to ensure transparency and accountability in the functioning of the AI.

Accountability can be defined as the ability to determine whether a decision was made in accordance with prescribed sets of standards and to hold liable a person if those standards are not met.¹⁴

Accountability in AI can be fixed by way of *Explanations*¹⁵. 'Explanation' can be defined as human-interpretable information about the logic as to how a decision maker took a particular set of inputs and reached a particular conclusion. Explanations can be used to rectify errors and to ascertain whether certain criteria were used appropriately in a dispute. A mechanism of 'Right to Explanation'¹⁶ for AI was debated at the meeting for the revision of European Union's General Data Protection Regulation (GDPR). In consequence of this, the revised GDPR effective May 2018 contains provision for 'Right to Information' in Article 13, Article 15 about existence, logic, consequences of automated decision making whereas Article 22 provides right to not be subjected to automated decision making process of AI¹⁷.

In the year 2016, a project was launched in the U.S names *Explainable AI* program to fund research into explainable AI models.¹⁸

Regulatory Framework of AI across Jurisdictions

European Union (EU):

The EU has developed *Directive on Copyright in Digital Single Market* 2019 (DSM Directive) for modernizing the Copyright law for digital age. Article 2 defines *Text and Data mining* clause which explicitly puts AI under

¹³ Hallevy, supra note 10

¹⁴ Joshua A Kroll, Solon Barocas, Edward W Felten, Joel R Reidenberg, David G Robinson, and Harlan Yu, *Accountable Algorithms*, 165 U.P.A.L.REV. 633, 656 (2016)

¹⁵ Finale Doshi-Velez & Mason Kortz, *Accountability of AI under the Law: The role of Explanation* (Berkman Klein Ctr. Working Grp. On Explanation & the Law, Working Paper, 2017)

¹⁶ Bryce Goodman and Seth Flaxman, *EU Regulations on Algorithmic decision-making and a 'Right to Explanation'*, AI MAGAZINE, Oct. 2017, at 50

¹⁷ Council Regulation 2016/679, arts. 13-15, 22, 2016 O.J. (L 119) 1

¹⁸ DARPA, Broad Agency Announcement, *Explainable Artificial Intelligence (XAI)*, DARPABAA-16-53, at 6, available at <https://www.darpa.mil/attachments/DARPABAA-16-53.pdf>

this definition. Whereas Article 4 gives the concept of ‘opt-out’, whereby reproduction for text and data mining is *permitted* unless right holders expressly reserve their rights. Article 17 which deals with *Platform/Intermediary liability* states that platforms can be made liable for the copyright violations on its platforms. This could potentially include to mean AI platforms as well.

EU does not confer Legal Personality to AI and therefore it is not Independently liable in Civil and Criminal cases, rather, liability is attached to human person behind the AI- it can be the manufacturer, importer, company using it etc.

An important piece of legislation governing AI is the *EU Artificial Intelligence Act 2024*¹⁹, which provides for a risk-based classification of AI. It divides AI into unacceptable, high risk, minimal and limited risk categories. Unacceptable risk AI are generally prohibited. This Act imposes duty on providers and deployers (large professional entities) of AI systems. There is a duty to maintain human oversight, maintain technical log of last 6 months etc. There is a provision of penalty of up to 35 million Pounds or 7 percent of the global turnover.

Some jurisdiction in EU have adopted the approach of assigning ‘*Strict liability*’ for AI systems meaning thereby preventing the developers, users, from pleading absence of intention, knowledge in committing the offence.

EU also has regulations like Digital Services Act²⁰, whereby online platforms have the liability to address the issues of unlawful content dissemination including through AI.

Countries in EU like Germany are proposing ethical guidelines for AI systems to ensure that such system have transparency, accountability. They also require developers to provide audit trails and documentation of such systems on regular basis.

United States of America (US):

Similar to the EU, US does not consider AI as a natural Independent legal person on which liability could be attached.

A major dispute regarding AI in Intellectual property is training of Large Language Models(LLMs) and AI systems by using copyright protected data. Such companies invoke the clause of ‘fair use’ under S-107 of Copyright Act 1976.

An important ongoing case in this regard is *The New York Times Company V. OpenAI and Microsoft*.²¹ This is one of the most important case regarding generative AI, copyright infringement and liability of developers. The plaintiff alleged that millions of copyrighted newspapers were used to train AI models like Chatgpt and Copilot without authorization. The issue here is that according to the plaintiff the reproduced work of the AI models closely resembles original work whereas the Defendant argue that this would come under ‘Fair Use’ and therefore they are protected from copyright infringement action. This matter is *subjudice* and could be a landmark to decide the future direction of such AI models. A similar case under consideration by the US court is related to unauthorized usage of millions of photographs to train AI system.²²

¹⁹ <https://digital-strategy.ec.europa.eu/en/policies/regulatory-framework-ai>, last visited on 01/05/2026

²⁰ <https://digital-strategy.ec.europa.eu/en/policies/digital-services-act>, last visited on 01/05/2026

²¹ *The New York Times Company v. Microsoft Corp. & OpenAI, Inc.*, No. 1:23-cv-11195(S.D.N.Y. filed Dec. 27,2023)

²² *Getty Images(US), Inc. v. Stability AI Ltd.* Claim No. IL-2023-000007

In the US, the approach adopted is *causation based* where criminal liability is attributed to human actor who made the AI commit a crime.²³

United Kingdom:

In a major landmark case²⁴, where a person developed an AI named DABUS and that AI independently and autonomously created two inventions. The plaintiff wanted to register the patent in the name of AI as it created both inventions but the registry refused it to register in the name of AI citing no law regarding it. The court relied on the statute UK Patents Act 1977, Section 7 and 13, according to which patent could be registered in the name of natural person and not AI. The Supreme Court rejected the contention of the plaintiff and held that since AI does not have Legal Personality therefore it can't be granted patent in its own name.

India:

The Copyright Act 1957, governs the Copyright related law in India. A major bone of contention is Section-2(d) which defines the term '*author*'. Firstly, the act states that only Natural person can be an author. So according to this definition autonomous AI can't be an author. Secondly, for computer-generated work, author is considered to be the person who created the work. As AI can't be considered to be the author under the present Law, the question arises whether it would be the end-user who gives the prompt or the developer who would be considered the author. Since the training of AI is done by using millions of documents, newspapers article, photographs which are protected works, this could be infringement of work punishable under Section 51. S-52 is the 'Fair Use' provision which is usually resorted to by the AI companies to avoid liability. They argue that AI training is transformative and research-oriented.

The criminal liability in India is governed by *Bhartiya Nyaya Sanhita* 2023, *Bhartiya Nagarika Suraksha Sanhita* 2023, and *Bhartiya Sakshya Adhiniyam*, 2023. As of now, there is no dedicated framework to govern AI liability in India. AI does not have legal personality in India and therefore it can't be treated as an Independent actor which could be made Independently liable. Therefore, at present, the liability could be attributed to the developers, operator, user, deployers.²⁵ Another important issue is the issue of Data privacy and security concerns. Hackers could misuse AI into giving them personal data of the users which is a major privacy and security concern. The Supreme Court of India has held in a landmark judgment that Right to Privacy is a Fundamental right ²⁶under Article 14,19,21 of the Indian Constitution and therefore it can't be infringed, the question which arises is what will happen when there are breaches of such sensitive user data by the AI. Although the present legal framework does not explicitly include AI in its ambit, but there are provisions in various Statutes under which it could be read into. For instance:

- Under the *Bhartiya Nyaya Sanhita*, 2023, Section 2(26) defines 'persons' which includes corporations but it does not include AI. S-111 addresses Organized crimes which also includes AI-driven frauds and Section 353 also covers misinformation.
- *Information and Technology Act 2000*: Section 66-D prescribes punishment which may extend to 3 years and 1 lakh Rupees fine in case of cheating by using computer resource. Section 66-E prescribes punishment for deep-fakes which carries imprisonment which may extent to 3 years and

²³ Kate Crawford & Jason Schultz, *AI Systems as State Actors*, 119 Column, L. Rev.1941(2019)

²⁴ Thaler v. Comptroller General of Patents, Designs and Trademarks (2023) UKSC 49

²⁵ Hifajatali Sayyed, *Artificial Intelligence and Criminal Liability in India: Exploring Legal Implications and challenges*, 10 Cogent Soc. Sci. No. 1, Art. 2343195(2024)

²⁶ *Justice K.S. Puttuswamy (Retd.) v. Union of India*, (2017) 10 SCC 1 (India)

fine may extend to 2 lakh rupees. These provisions cover identity theft, impersonation, AI-generated content within its ambit.

- Information Technology (Intermediary guidelines and Digital media ethics Code) Rules 2021: These rules make intermediary liable for the content they host and therefore plugs a key loophole in governance.
- A February 2026 amendment ²⁷has taken place to the IT rules 2021 whereby the government has shifted its approach from reactive to proactive by shifting the liability on the intermediary to regulate the content on their platform which include deep-fakes fake audio clips. The regulation requires the mandatory takedown of deep-fakes within 3 hours of serving notice (earlier it was 36 hours). It has been mandated that platforms use some technical markers that will help in tracing AI generate content to its original source.

A bill²⁸ was introduced in Indian Parliament by a Private member which sought to regulate the AI and make it accountable. *Artificial Intelligence (Ethics and Accountability) Bill 2025* sought to establish India's first dedicated law on AI to regulated ethical development, governance of AI. The bill seeks to establish an Ethics Committee which will see the governance of the AI. The responsibility is imposed on the Developers, Deployers, Companies for negligence of the AI for instance deep-fakes etc. It also seeks to deploy mechanisms like mandatory audits, risk assessment, grievance redressal mechanisms.

Nakul Bajaj case ²⁹is a famous case regarding Intermediary liability whose application could be extended to AI platforms as well. Plaintiff sued an online platform which allegedly sold counterfeit products using Plaintiff's trademark. The defendant claimed Intermediary protection under the Information Technology Act 2000. The Delhi High Court held that platform that exercise significant control over content and transactions cannot claim Safe harbour protection. This principle could also be extended to AI platforms and can't avoid liability by claiming that AI autonomously generated IP protected content.

Another landmark case in this regard is Shreya Singhal's case ³⁰where the Constitutionality of Section-66A, which punished sending offensive messages electronically and Section 79 which deals with Intermediary Liability. The Supreme Court struck down Section 66-A for being too vague in its terminology, and for Intermediary liability *the court held that they can't be made automatically liable for third-party content*. They are required to remove content when ordered by a court or by government authority.

The primary objective of this study is to critically examine the attribution of criminal liability in AI-enabled technology-related offences, with particular emphasis on intellectual property violations. The research evaluates the adequacy of existing criminal and intellectual property laws in addressing offences committed through AI systems and analyses comparative legal approaches adopted in India, the European Union, the United States, and the United Kingdom. It further seeks to identify regulatory deficiencies and propose a balanced accountability framework that safeguards innovation while ensuring effective legal enforcement through mechanisms such as strict liability, vicarious liability, algorithmic transparency, audit trails, explain ability, and human oversight.

Existing scholarship has extensively examined the implications of AI for criminal liability and intellectual property rights. Scholars such as John McCarthy conceptualized AI as the science of creating intelligent machines, while Gabriel Hallevy proposed theoretical models for attributing criminal liability to AI, including the "perpetration-via-another" and direct liability approaches. Research by Joshua Kroll and

²⁷ <https://www.livewlaw.in/articles/ai-generated-content-deepfakes-524064>, last visited on 01/05/2026

²⁸ <https://www.sconline.com/blog/post/2025/12/19/artificial-intelligence-ethics-accountability-bill-2025-legal-update>, last visited on 05/05/2026

²⁹ Christian Louboutin SAS V. Nakul Bajaj, 2018 SCC Online Del. 12915

³⁰ Shreya Singhal v. Union of India, (2015) 5 SCC 1

colleagues has emphasized algorithmic accountability and transparency, whereas Goodman and Flaxman highlighted the emerging "right to explanation" within AI governance.

Recent literature has also analysed regulatory initiatives such as the European Union Artificial Intelligence Act, the Digital Services Act, and ongoing copyright litigation involving generative AI systems in the United States. Although these studies significantly contribute to the discourse on AI governance, much of the existing research remains fragmented, focusing either on intellectual property law or criminal liability independently. Limited attention has been paid to integrating technological realities with criminal accountability across multiple stakeholders and jurisdictions. This study addresses that gap by adopting a comprehensive doctrinal and comparative approach to develop a practical framework for regulating AI-enabled offences in the evolving digital ecosystem.

2. Methodology

- **Research Design**

This study adopts a qualitative, doctrinal, comparative, and analytical research design. The doctrinal approach is employed to examine the existing legal framework governing Artificial Intelligence (AI), intellectual property (IP), and criminal liability by analysing statutory provisions, judicial precedents, international conventions, policy documents, and scholarly literature. The comparative method evaluates regulatory approaches adopted in India, the European Union, the United States, and the United Kingdom to identify similarities, differences, and emerging best practices for AI governance. The analytical approach is used to assess the adequacy of existing legal principles, particularly *actus reus*, *mens rea*, legal personality, strict liability, and vicarious liability, in addressing AI-enabled technology-related offences.

- **Data Sources (Participants/Sample)**

As this is a non-empirical legal study, no human participants or statistical sample were involved. Instead, the research is based on purposively selected primary and secondary legal sources.

The primary sources include constitutional provisions, criminal statutes, intellectual property legislation, information technology laws, judicial decisions, government reports, policy documents, and regulatory frameworks relating to AI from India, the European Union, the United States, and the United Kingdom. Secondary sources comprise peer-reviewed journal articles, books, reports published by international organisations, law commission reports, conference papers, and recent news reports relating to AI-enabled crimes and intellectual property infringement.

- **Research Instruments**

The principal research instruments consisted of document analysis and legal content analysis. Statutory provisions, judicial precedents, policy documents, and academic literature were systematically reviewed to identify legal principles governing AI accountability. Online legal databases, official government websites, international institutional reports, and peer-reviewed legal literature were utilised to collect and verify relevant information. Comparative legal analysis was undertaken to examine legislative developments and judicial approaches across selected jurisdictions.

• **Research Procedure**

The research was conducted through a structured five-stage process. First, relevant literature relating to Artificial Intelligence, criminal liability, and intellectual property law was reviewed to identify existing debates and research gaps. Second, primary legal materials, including statutes, judicial decisions, and policy frameworks from India and selected foreign jurisdictions, were collected and examined. Third, significant AI-enabled offences, including deep fake fraud, cybercrime, automated piracy, copyright infringement, and identity theft, were analysed through contemporary case studies and legal developments. Fourth, comparative analysis was undertaken to evaluate the regulatory approaches adopted by different jurisdictions towards AI governance and liability attribution. Finally, the findings were critically analysed to develop recommendations for strengthening accountability through hybrid liability models incorporating strict liability, algorithmic transparency, audit mechanisms, explain ability, and human oversight.

• **Data Analysis**

The collected materials were analysed using qualitative doctrinal and comparative legal analysis. Statutory interpretation, case law analysis, and thematic examination of legal literature were employed to evaluate the effectiveness of existing legal frameworks. Comparative analysis enabled identification of regulatory trends, legislative gaps, and best practices across jurisdictions. The findings were synthesised to assess the adequacy of current criminal and intellectual property laws in addressing AI-enabled technology-related offences and to formulate recommendations for an adaptive and internationally coordinated framework for AI accountability.

3. Results

The doctrinal and comparative analysis reveals significant legal and regulatory gaps in attributing criminal liability for Artificial Intelligence (AI)-enabled technology-related offences. Existing criminal law frameworks across major jurisdictions continue to treat AI as a technological instrument rather than an independent legal entity, resulting in uncertainty regarding the attribution of liability where autonomous or semi-autonomous AI systems contribute to criminal conduct. Comparative examination further demonstrates that although jurisdictions have adopted varying regulatory approaches, no jurisdiction presently recognises AI as a legal person capable of bearing independent criminal liability.

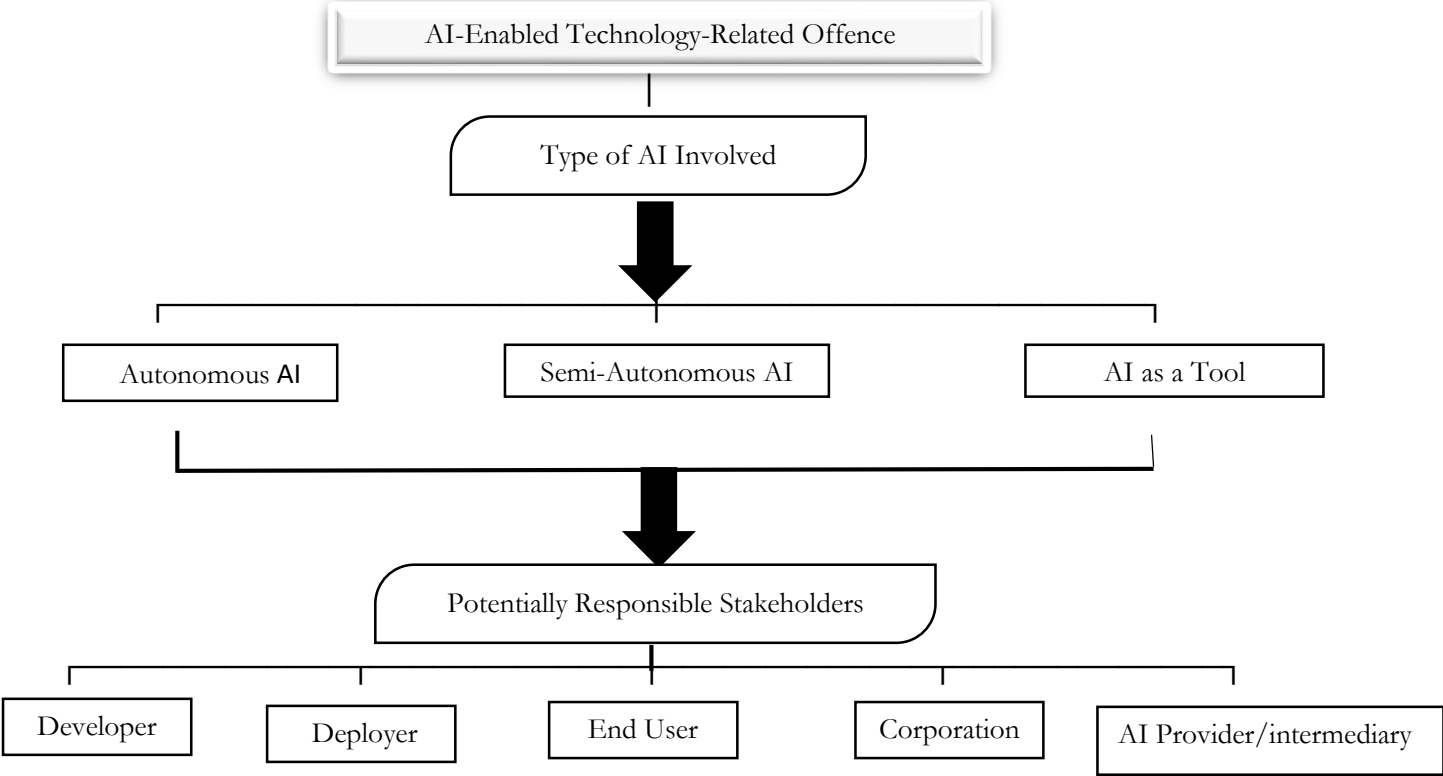
Table 1. Comparative Legal Position on AI Criminal Liability

Jurisdiction	Legal Status of AI	Primary Liability	Key Regulatory Framework
India	No legal personality	Developer, user, intermediary, deployer	Bharatiya Nyaya Sanhita, 2023; Information Technology Act, 2000; IT Rules, 2021
European Union	No legal personality	Providers, manufacturers, deployers	AI Act, 2024; Digital Services Act; DSM Directive, 2019
United States	No legal personality	Human actor based on causation	Copyright Act, 1976; State and federal laws
United Kingdom	No legal personality	Human applicant/developer	Patents Act, 1977; judicial precedents

Table 2. Major Findings of the Study

Finding No.	Observation
F1	Existing criminal law is based upon <i>actus reus</i> and <i>mens rea</i> , making attribution of criminal intent to AI difficult.
F2	AI presently lacks legal personality in all jurisdictions examined.
F3	Liability is generally imposed upon developers, deployers, users, corporations, or intermediaries rather than AI itself.
F4	Existing intellectual property laws inadequately regulate AI-generated copyright and trademark infringement.
F5	Regulatory frameworks differ considerably among jurisdictions, resulting in fragmented governance.
F6	Explain ability, audit mechanisms, transparency, and human oversight are increasingly incorporated into AI governance frameworks.
F7	No comprehensive international legal framework presently governs AI criminal liability.
F8	Hybrid liability models combining strict liability, vicarious liability, and enterprise liability are emerging as potential regulatory approaches.

Figure 1. Stakeholders Potentially Responsible for AI-Enabled Offences



Principal Findings

The review of statutory provisions, judicial decisions, and policy frameworks indicates that no jurisdiction presently confers independent criminal legal personality upon AI systems. Consequently, criminal liability continues to be attributed to human or corporate actors associated with the development, deployment, or operation of AI technologies.

The comparative analysis further demonstrates that the European Union has established the most comprehensive regulatory framework through risk-based governance under the Artificial Intelligence Act, 2024, whereas India primarily regulates AI indirectly through existing criminal, intellectual property, and information technology laws. The United States continues to address AI-related disputes largely through existing copyright doctrines and judicial interpretation, while the United Kingdom maintains that AI cannot independently qualify as an inventor or legal person.

The analysis also identifies increasing judicial and legislative concern regarding AI-enabled offences including deep fake fraud, automated cyberattacks, identity theft, copyright infringement, trademark misuse, and algorithmic decision-making. Existing legal frameworks predominantly rely upon traditional concepts of human responsibility, resulting in regulatory uncertainty where autonomous AI systems operate with limited human intervention.

Across all jurisdictions examined, regulatory developments increasingly emphasise transparency, algorithmic accountability, audit requirements, explain ability, risk assessment, and human oversight as essential governance mechanisms for AI systems.

4. Discussion

Proposed reforms

- The challenges posed by the AI can be tackled by considering conferring *Legal Personality* to AI like the Corporations so that civil and criminal liability could be conferred on the AI.
 - In order to hold AI liable for AI-related offences, the concept of '*Strict Liability*' should be introduced like it is introduced in some European countries, so that AI-developers and deployers can be held liable for such offences irrespective of Intent or Knowledge. By this it could be ensured that at the development stage, there would be responsible development of AI which would lead to more transparency and accountability.
 - It should also be mandated that AI developers file clear *documentation and audit trails* of the AI systems so that there could be better understanding as to the decision making process of the AI system which would lead to a more transparent and accountable system.
 - Another important change that could be made is to promote *framework at National and International level* that seeks to enhance transparency and *Explain ability* of the AI-system.
 - Since the domain of such AI activity is in many cases International and cross-border, therefore there is a need for *Unified Regulatory structure* across the world so that regulation of AI would become seamless and unified. International organizations like United Nations, G-7, G-20, BRICS, ASEAN, SCO could become the leaders by effectively guiding the nations in seamless transition to the AI world.
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- There could be a AI-specific *AI Data Protection Act* to regulate AI and create liability of it.
- It is proposed by research scholars that the frameworks should be developed so as to ensure that *humans have the ultimate control* (humans-in-the-loop) over critical AI decisions, for instance, in automated policing systems³¹.
- AI systems ³²are vulnerable to cyberattacks and this could be exploited by the hackers who operate at global level. To ensure that sensitive data used in AI training and LLMs are not misused it is to be ensured that the data of AI is secure with proper mechanisms in place. In addition, it should be made mandatory by law to report any misuse of AI in deep-fakes, automated frauds.
- Efforts are need to be made to make AI accountable during the design stage itself by preventing biasness in programming of AI, ensuring privacy and ethical governance principles.
- The laws governing Intellectual Property could be appropriately amended to accommodate AI into its framework.

Limitation

This study is primarily a doctrinal and comparative legal research which focus over the criminal liability of AI. Therefore, it does not include empirical research like surveys, interviews or stakeholder interactions with AI developers, law enforcement agencies, policymakers, or judicial officers. Also AI is into peak of its development and it is still in development phase which can confer to new bills, proposed legislature or different statutory approach for the same hence this research is limited to only current scenario and is unable to address the future situations relating to criminal AI accountability.

This study is confined to only limited jurisdiction as comparative analysis like USA, UK, European countries and India. It does not compare any other countries jurisdiction for the study.

5. Conclusion

AI is our future and it is better if we accept this fact and try to work our way through the challenges put forth by the AI. Since it is cross-border and global in nature, it is high time we devised frameworks that is global in nature and which puts forth a *Unified Regulatory Framework* for the world which each member countries could adopt into their domestic law, just like IP framework. A major issue with AI is that the structure of AI works as '*Black-box*' i.e. it is very difficult to know how the system has arrived at a particular

³¹ Laxmiraju Kandikatla & Branislav Radeljic, *AI and Human Oversight: A risk-based framework for Alignment* (Oct. 10, 2025)

³² Rowena Rodrigues, *Legal and Human Rights Issues of AI: Gaps, challenges and vulnerabilities*, 4 J. Responsible Tech. 100005(2020)

conclusion or output. To solve this issue, a *hybrid-accountability* model whereby regular audit, compliance mechanism could be put in place so that this issue could be dealt with. Present Criminal Law and IP framework does not explicitly mention AI in its provisions and therefore these laws have to be suitably amended. EU is leading the world in this regard where dedicated AI regulations and legal frameworks are put in place. Other countries should follow suit.

The AI has opened the doors of the future with unlimited possibilities and opportunities. It is up to us to make use of it in the best possible way while efficiently evolving our legal framework to deal with the emerging issues. For this, the world needs to work as one family.

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8. Author Declaration

The authors declare that this manuscript is an original work and has not been published previously, nor is it under consideration for publication elsewhere, in whole or in part. The research has been conducted in accordance with accepted principles of academic integrity, research ethics, and scholarly practice. All sources of information have been appropriately acknowledged and cited.

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